Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/04/2011
PI Name:	Fike, John R. Ph.D.		
Project Title:	Use of a molecular marker of learning and memory to assess effects of 56Fe irradiation on hippocampus-dependent cognition and neurogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94110	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Raber, Jacob (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
Grant/Contract No.:	NNX08BA07G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Particulate irradiation in the space environment may pose a significant risk to space flight crews during a mission or upon return to Earth. This may involve the central nervous system, where low radiation doses can induce cognitive or behavioral impairments. Understanding the cellular and molecular basis of cognitive injury will facilitate the establishment of risk estimates and may provide a foundation for the establishment of effective countermeasures or treatments for such injury. Using an animal model, we will determine the extent to which low doses of 56Fe irradiation affect cognitive function and will determine if such changes are associated with the alteration of a gene that is associated with learning and memory, or with the production and maturation of newly born neurons.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The chosen animal model and the behavioral, cellular and molecular endpoints to be used will provide new insight regarding how high LET irradiation affects factors associated with cognitive or behavioral function. Further, the design of these studies, using relatively low doses such as those that could be experienced in a long term space mission, will allow us to determine if there is a threshold of such effects. This could have significant impact in the determination of risk for astronauts during a prolonged mission. Additionally, understanding the association between neurogenesis, the expression of the immediate early gene Arc, and behavioral performance, will provide valuable new information about the how cognitive function is impacting by deleterious insults such as irradiation. Such information could have an impact in a number of biomedical fields including radiation oncology, neurodegeneration and cognitive neuroscience.
Task Progress:	The overall goal of this project is to determine how low doses of 56Fe ions affect behavioral performance and specific cellular and molecular factors that may play a role in cognition. Additionally, it is an objective of this project to determine if/how those various endpoints are affected by inflammatory changes. The experimental methods chosen to accomplish these goals provide a means by which the effects of low radiation doses can be quantified both in terms of behavioral performance as well as specific cellular and molecular endpoints that are associated with learning and memory. In the current funding period we irradiated C57BL6 wild type (WT) mice (n = 120) and animals deficient in chemokine receptor-2 (CCR2 knock out — KO mice, n = 100) with 0, 0.25 and 0.5 Gy of 56Fe. We also finalized our analyses of WT mice irradiated in Year 2 with brain only doses of 1 Gy. Mice were randomly assigned to the various treatment cohorts and were group housed until 60 days after irradiation when they were singly housed for the duration of the experiment. The mice from treatment cohort were randomly assigned to 5 experimental groups of 8 mice each. Groups 1 and 2 underwent behavioral testing in a contextual fear conditioning paradigm (hippocampus-dependent) and Groups 3 and 4 underwent behavioral testing in a cued fear conditioning paradigm (hippocampus-independent). Group 5 constituted caged controls and did not undergo any behavioral testing. For our assessments of behavior we used a fear conditioning testing paradigm that consist of two versions, one that requires the hippocampus (context version) and one that does not (cued version). Both versions involve, in part, the amygdala, a brain region associated with fear and anxiety, and that to our knowledge has not been well-studied in terms of radiation response. In the fear conditioning tests that were used, mice had to learn to associate a neutral stimulus, i.e. a tone (CS), with a foot shock (US), and thereby come to fear the CS. Trained mice displayed this conditioned fear by ceasing all movement except for respiration in an attitude called 'freezing.' Training took place in a light and sound attenuated chamber (termed the 'conditioning chamber') that was equipped with a video camera to record freezing behavior. The neural circuitry underlying fear conditioning is well documented and so this type of learning can be used to assess the functioning of discrete brain regions in mouse models of neurological disease or injury. The type of training determines the brain regions that are involved in the learning and memory processes. On the first day of testing, after a 2 min period, during which the mouse was allowed to explore the chamber, a tone was delivered followed after 30 sec by a foot shock (0.35 mA over 2 seconds). Tests for learning occurred 24 hours after training, and took place in either the conditioning chamber or a different control chamber (termed the 'non-conditioning chamber'). Thirty minutes after the end of the final testing session, mice were killed by cervical dislocation, decapitated, and the brains were prepared for microscopic analyses. We completed all analyses of cognitive performance and of the expression of the behaviorally induced immediate early gene Arc in animals that received brain only irradiation. Irradiated mice that did not receive tone-shock pairings on Day 1 showed less freezing in the same context on a second day and a lower fraction of Arc-expressing neurons in the free (lower) blade of the DG than sham-irradiated mice. Those data suggested reduced hippocampus-dependent spatial habituation learning. Changes in Arc expression in the free blade correlated positively with freezing in mice that did not receive tone shock pairings. However, irradiated mice that did receive tone shock pairings showed enhanced contextual freezing but a reduced percentage of Arc-expressing neurons in the enclosed (upper) blade. Changes in Arc expression correlated negatively with freezing in mice that received tone shock pairings. In animals receiving cued fear conditioning, irradiation did not affect cognitive performance or the fractions of Arc-expressing neurons. While the relationship between Arc expression and cognitive performance is complex, our data suggest that radiation effects on hippocampus-dependent cognition might depend on the prominence (salience) of environmental stimuli and blade-specific Arc expression. In our studies of low-dose whole body irradiation, we completed the analysis of behavioral performance and measures of neurogenesis (newly born neurons); analysis of Arc expression is currently underway. These studies showed that irradiated wild type (WT) mice showed more habituation to the environment, than sham irradiated animals, and that the effect was dose dependent. Compared to WT mice, animals deficient in CCR2, showed less freezing or more exploratory activity than WT mice. Additionally, CCR2 KO mice that received a dose of 0.25 Gy did not show the same response as WT and had similar freezing as genotype matched sham irradiated mice. This indicated that after the lower dose, CCR2 KO mice showed an impaired hippocampus-dependent spatial habituation. The responses seen after tone shock pairing were very different with both non-irradiated WT and KO mice showing robust freezing when placed in the adverse environment for the second time. After a dose of 0.25 Gy there was less freezing in CCR2 KO mice and a trend toward less freezing in WT mice, suggesting impaired hippocampal function after this dose, and increased susceptibility of CCR2 KO mice to this effect. It is particularly interesting that this was not seen after 0.5 Gy, suggesting that the effects of relatively low radiation doses on hippocampus dependent function are complex and critically dependant upon dose. Cellular analyses of newly born neurons showed that neurogenesis was reduced in only one group, the irradiated CCR2 KO mice. This result was consistent with the tone-shock pairing results described above. Our data, while not yet complete, highlight the complexity of assessing how low doses of 56Fe affect hippocampal function, and if those effects are impacted by an altered microenvironment (e.g. CCR2 deficiency). While the mechanisms responsible are still not yet known, our data suggest that even a relatively low dose of 56Fe can induce behavioral impairments that involve the hippocampus. If or how those impairments involve the process of neurogenesis, the expression of the behaviorally induced immediate early gene Arc, and the presence or absence of neuroinflammation need to be further investigated.
Bibliography Type:	Description: (Last Updated: 04/11/2018)

Articles in Peer-reviewed Journals	Raber J, Rosi S, Chakraborti A, Fishman K, Dayger C, Davis M, Villasana L, Fike JR. "Effects of 56Fe cranial radiation on hippocampus-dependent cognition and Arc protein expression in the dentate gyrus depend on the salience of the environmental stimuli." Radiation Research, In Press, as of August 2011. , Aug-2011